Exact Deficit Identity on Reduced-Boundary Level Sets (Route δ, Plan 2, Building Block LevelSetIdentity)

Abstract

We prove that for every open set $\Omega \subset \mathbb{R}^n$ of finite measure, with $u = u_\Omega \in H_0^1(\Omega)$ the variational torsion function and $E_t := \{u > t\}$, $m(t) := |E_t|$, the Saint–Venant deficit admits the exact representation

$$\frac{2}{|\Omega|}\big(E(\Omega) - E(B)\big) = \frac{1}{|\Omega|}\int_0^{\|u\|_\infty} \frac{D_H(t) + D_I(t)}{n^2\omega_n^{2/n}\, m(t)^{1-2/n}}\, dt,$$

where B is the ball with $|B| = |\Omega|$, $D_H(t)$ is the Cauchy–Schwarz defect of $|\nabla u|$ on the reduced boundary $\partial^* E_t$, and $D_I(t)$ is the squared-perimeter isoperimetric defect of E_t. The proof rests on four ingredients which all hold on reduced boundaries for a.e. level: (i) the BV coarea identity for $-m'(t)$, (ii) a weak divergence-flux identity proved by Lipschitz truncation, (iii) the distributional second derivative of the rearranged primitive $I(s) = \int_0^s u^*$, in the corrected form $I'' = -1/\alpha$ with $\alpha = \int_{\partial^* E_t} |\nabla u|^{-1}$, and (iv) the endpoint convexity gap together with the Pólya–Szegő identification of the ball energy. No graph regularity, no BDV selection, no smoothness of $\partial\Omega$, and no boundedness or connectedness of Ω are used. The identity is the foundational input for the downstream Plan 2 blocks [14, 16, 17, 18, 19].

1 Notation and standing hypotheses

Throughout $n \geq 2$. We fix the following objects and use them without further comment.

- $\Omega \subset \mathbb{R}^n$ open with $|\Omega| < \infty$. No connectedness, smoothness, or boundedness is assumed.
- $u = u_\Omega \in H_0^1(\Omega)$ is the variational torsion function, i.e. the unique minimizer of $J(v) = \frac{1}{2}\int |\nabla v|^2 - \int v$ over $H_0^1(\Omega)$; equivalently the weak solution of $-\Delta u = 1$ in Ω, $u = 0$ on $\partial\Omega$ (in the H_0^1 trace sense). Testing against u^- yields $u \geq 0$ a.e. by the weak maximum principle.
- $E(\Omega) := -\frac{1}{2}\int_\Omega u\, dx = -\frac{1}{2}T(\Omega)$, where $T(\Omega)$ is the torsional rigidity.
- $B \subset \mathbb{R}^n$ is the open ball with $|B| = |\Omega|$, radius $R := (|\Omega|/\omega_n)^{1/n}$, $\omega_n = |B_1|$. The function u_B is the torsion function of B.
- For $t \geq 0$, $E_t := \{u > t\}$ (the precise representative as in [11, §15]), $m(t) := |E_t|$. The function m is right-continuous, nonincreasing, with $m(0^+) \leq |\Omega|$ and $m(\|u\|_\infty) = 0$.
- $u^* \colon [0, |\Omega|] \to [0, \|u\|_\infty]$ is the decreasing rearrangement, $u^*(s) := \inf\{t \geq 0 : m(t) \leq s\}$, in the convention of [7, 13].
- $\partial^* E_t$ denotes the De Giorgi reduced boundary, with $\mathcal{H}^{n-1}(\partial E_t \setminus \partial^* E_t) = 0$ for a.e. t [5].
- We set
$$\alpha(t) := \int_{\partial^* E_t} |\nabla u|^{-1}\, d\mathcal{H}^{n-1}, \qquad \beta(t) := \int_{\partial^* E_t} |\nabla u|\, d\mathcal{H}^{n-1},$$
defined for a.e. t (see Lemmas 2.1 and 3.1), and $c_n := n^2\omega_n^{2/n}$.

Talenti L^∞ bound. By the Talenti pointwise comparison [13],

$$u^*_\Omega(s) \le u^*_B(s) = \frac{R^2 - (s/\omega_n)^{2/n}}{2n}, \qquad s \in [0, |\Omega|], \tag{1}$$

so $\|u\|_{L^\infty(\Omega)} \le R^2/(2n) < \infty$. This is the only reason boundedness of Ω is not needed at the analytic level.

Coarea applicability. Since $u \in H^1_0(\Omega) \subset BV_{\mathrm{loc}}(\mathbb{R}^n)$ (extended by 0) and $u \in L^\infty$, the BV coarea formula [11, Thm. 13.1], [1, Thm. 3.40] gives, for every Borel $\varphi \ge 0$,

$$\int_\Omega |\nabla u|\, \varphi \, dx = \int_{-\infty}^{\infty} \Big(\int_{\partial^* E_t} \varphi \, d\mathcal{H}^{n-1} \Big)\, dt. \tag{2}$$

The set

$$\mathcal{R} := \big\{ t \in (0, \|u\|_\infty) : P(E_t) < \infty,\ \partial^* E_t \subset \Omega \big\}$$

has full Lebesgue measure in $(0, \|u\|_\infty)$; Sobolev–Sard [6, App. A, Thm. A.1] additionally gives $\mathcal{H}^{n-1}(\{u = t\} \cap \{|\nabla u| = 0\}) = 0$ for a.e. t, but is not used below since all integrals are on reduced boundaries. Statements "for a.e. t" below refer to a.e. $t \in \mathcal{R}$.

2 Coarea identity for $-m'(t)$

Lemma 2.1 (Coarea form of $-m'$)**.** *For a.e. $t \in (0, \|u\|_\infty)$,*

$$-m'(t) = \int_{\partial^* E_t} \frac{1}{|\nabla u|}\, d\mathcal{H}^{n-1} = \alpha(t). \tag{3}$$

In particular m is absolutely continuous on $[\varepsilon, \|u\|_\infty]$ for every $\varepsilon > 0$, and $\alpha \in L^1_{\mathrm{loc}}((0, \|u\|_\infty))$.

Proof. Fix $0 < \alpha_0 < \beta_0 < \|u\|_\infty$ and apply (2) to the Borel function $\varphi = |\nabla u|^{-1} \mathbf{1}_{\{|\nabla u| > 0\}} \mathbf{1}_{[\alpha_0, \beta_0]}(u)$. By [11, Thm. 13.1],

$$\int_\Omega \mathbf{1}_{\{\alpha_0 < u < \beta_0\}}\, dx = \int_{\alpha_0}^{\beta_0} \Big(\int_{\partial^* E_t} |\nabla u|^{-1}\, d\mathcal{H}^{n-1} \Big) dt,$$

because the integrand on the left equals $|\nabla u| \cdot |\nabla u|^{-1} \mathbf{1}_{\{|\nabla u|>0\}} \mathbf{1}_{\{\alpha_0 < u < \beta_0\}}$ and the contribution of $\{|\nabla u| = 0\}$ to the BV coarea slicing vanishes [1, 3.40(b)]. The left-hand side equals $m(\alpha_0) - m(\beta_0)$ outside the at most countably many plateau levels of u (levels t with $|\{u = t\}| > 0$; each such plateau contributes a jump of m, and m is monotone, so these are at most countably many). Differentiating in β_0 at a.e. β_0 yields (3). $\square$

Remark 2.2. At the countable set of plateau levels (3) is not asserted; the Lebesgue integral in the main theorem absorbs them trivially.

3 Weak divergence-flux identity

Lemma 3.1 (Flux = volume)**.** *For a.e. $t \in (0, \|u\|_\infty)$, E_t has finite perimeter and*

$$\beta(t) = \int_{\partial^* E_t} |\nabla u|\, d\mathcal{H}^{n-1} = m(t). \tag{4}$$

Proof. Fix $t \in \mathcal{R}$ with $|E_t| < \infty$, $P(E_t) < \infty$, and u not having a plateau at t (all these hold for a.e. t).

Step 1 (Lipschitz truncation). For each $\varepsilon > 0$ set

$$\phi_\varepsilon(s) := \min\bigl(1, \max(0, s/\varepsilon)\bigr), \qquad \eta_\varepsilon(x) := \phi_\varepsilon\bigl(u(x) - t\bigr).$$

Since ϕ_ε is Lipschitz with $\phi_\varepsilon(0) = 0$ and $u \in H^1_0(\Omega)$, one has $\eta_\varepsilon \in H^1_0(\Omega)$. Testing the weak equation $-\Delta u = 1$ in $H^1_0(\Omega)$ against η_ε gives

$$\int_\Omega \nabla u \cdot \nabla \eta_\varepsilon \, dx = \int_\Omega \eta_\varepsilon \, dx. \tag{5}$$

The chain rule for Sobolev compositions [1, Thm. 3.99] gives $\nabla \eta_\varepsilon = \varepsilon^{-1} \nabla u \, \mathbf{1}_{\{t<u<t+\varepsilon\}}$, so

$$\int_\Omega \nabla u \cdot \nabla \eta_\varepsilon \, dx = \frac{1}{\varepsilon} \int_{\{t<u<t+\varepsilon\}} |\nabla u|^2 \, dx.$$

Step 2 (Coarea on the right of (5)). Apply (2) with $\varphi = |\nabla u| \, \mathbf{1}_{(t,t+\varepsilon)} \circ u$:

$$\int_{\{t<u<t+\varepsilon\}} |\nabla u|^2 \, dx = \int_t^{t+\varepsilon} \Bigl(\int_{\partial^* E_\tau} |\nabla u| \, d\mathcal{H}^{n-1} \Bigr) d\tau = \int_t^{t+\varepsilon} \beta(\tau) \, d\tau.$$

Step 3 (Limit on the left of (5)). $\eta_\varepsilon \to \mathbf{1}_{E_t}$ pointwise as $\varepsilon \downarrow 0$ (away from the level set $\{u = t\}$, which has Lebesgue measure zero since t is not a plateau) and $0 \le \eta_\varepsilon \le 1$, so by dominated convergence $\int_\Omega \eta_\varepsilon \to |E_t| = m(t)$.

Step 4 (Limit on the right). The map $\tau \mapsto \beta(\tau)$ belongs to $L^1_{\mathrm{loc}}((0, \|u\|_\infty))$: by (2) with $\varphi = |\nabla u| \, \mathbf{1}_{(0,\|u\|_\infty)} \circ u$,

$$\int_0^{\|u\|_\infty} \beta(\tau) \, d\tau = \int_\Omega |\nabla u|^2 \, dx = \int_\Omega u \, dx < \infty,$$

the last identity being the Dirichlet identity obtained by testing $-\Delta u = 1$ against $u \in H^1_0(\Omega)$. At every Lebesgue point of β, hence for a.e. t,

$$\frac{1}{\varepsilon} \int_t^{t+\varepsilon} \beta(\tau) \, d\tau \xrightarrow[\varepsilon \downarrow 0]{} \beta(t).$$

Step 5. Combining Steps 1–4 with (5) yields $\beta(t) = m(t)$ for a.e. t. ☐

Remark 3.2 (Formal divergence-theorem reading)**.** The argument of Lemma 3.1 is the rigorous BV/Lipschitz-truncation replacement for the formal divergence theorem on E_t: $\nabla u \cdot \nu_{E_t} = -|\nabla u|$ on $\partial^* E_t$ (since $u > t$ inside), so $-\Delta u = 1$ would give

$$m(t) = \int_{E_t} 1 \, dx = -\int_{E_t} \Delta u \, dx = -\int_{\partial^* E_t} \nabla u \cdot \nu_{E_t} \, d\mathcal{H}^{n-1} = \beta(t).$$

Without a boundary trace of ∇u on $\partial^* E_t$, the Lipschitz truncation above is what makes this rigorous on arbitrary finite-perimeter level sets [11, Ch. 17].

Remark 3.3 (Integrated Dirichlet identity)**.** Integrating (4) against dt and using (2) recovers the Dirichlet identity $\int_0^{\|u\|_\infty} \beta(t) \, dt = \int_\Omega |\nabla u|^2 = \int_\Omega u = 2|E(\Omega)|$ used above. Lemma 3.1 is its pointwise level-wise version.

4 Distributional derivatives of the rearranged primitive

Define

$$I(s) := \int_{\{u>u^*(s)\}} u\,dx, \qquad s \in [0,|\Omega|]. \tag{6}$$

The following three lemmas identify I with a one-dimensional primitive of u^* and compute its second distributional derivative in the form used by the deficit identity.

Lemma 4.1 (Layer-cake form of I)**.** *For every $s \in [0,|\Omega|]$,*

$$I(s) = \int_0^s u^*(\sigma)\,d\sigma. \tag{7}$$

Proof. By equimeasurability of u and u^* on $(0,|\Omega|)$ [7, §2], [13], $\int_\Omega g(u)\,dx = \int_0^{|\Omega|} g(u^*(\sigma))\,d\sigma$ for every Borel $g \geq 0$. Choose $g(t) = t\,\mathbf{1}_{\{t>u^*(s)\}}$; since $\{\sigma \in (0,|\Omega|) : u^*(\sigma) > u^*(s)\} = (0,s)$ up to null sets by the very definition of u^*, we obtain

$$I(s) = \int_\Omega g(u)\,dx = \int_0^{|\Omega|} g(u^*(\sigma))\,d\sigma = \int_0^s u^*(\sigma)\,d\sigma.$$

□

Lemma 4.2 (First derivative)**.** *I is absolutely continuous on $[0,|\Omega|]$ and $I'(s) = u^*(s)$ for a.e. $s \in (0,|\Omega|)$.*

Proof. By (1) $u^* \in L^\infty(0,|\Omega|)$; in particular u^* is bounded and Borel, hence in $L^1(0,|\Omega|)$ (since $|\Omega| < \infty$). By Lemma 4.1, I is the indefinite integral of u^*, so the Lebesgue differentiation theorem gives the claim. □

Lemma 4.3 (Second derivative — corrected form)**.** *The rearrangement u^* is absolutely continuous on $[s_0,|\Omega|-\varepsilon]$ for every $0 < s_0 < |\Omega|-\varepsilon < |\Omega|$, and*

Two intermediate steps justify the displayed chain below: (a) the identity $I'' = (u^*)'$ follows from Lemma 4.2 ($I' = u^*$) combined with the absolute continuity of u^* on closed subintervals of $(0,|\Omega|)$ (so that the distributional derivative of I' coincides with the a.e. classical derivative of u^*); and (b) $\alpha(t) > 0$ for a.e. t, since $\alpha = -m'$ by Lemma 2.1 and m is strictly decreasing a.e. on its non-plateau range, so the reciprocal is well defined a.e.

$$I''(s) = (u^*)'(s) = \frac{1}{m'(u^*(s))} = -\frac{1}{\alpha(u^*(s))} \qquad \text{for a.e. } s \in (0,|\Omega|). \tag{8}$$

Proof. m is right-continuous, nonincreasing, and (Lemma 2.1) absolutely continuous on $[\varepsilon,\|u\|_\infty]$ for every $\varepsilon > 0$, with $m'(t) = -\alpha(t)$ a.e. The plateau levels of u (countable, by monotonicity of m) are the levels where m jumps; outside them, m is strictly decreasing on its domain of strict monotonicity, hence its right-continuous generalized inverse u^* is also strictly decreasing and absolutely continuous on closed subintervals of $(0,|\Omega|)$. Differentiating $m(u^*(s)) = s$ at a.e. s via the inverse-function theorem for monotone AC functions [1, Thm. 3.99], [3] gives (8). □

Remark 4.4 (Notational fix vs. the source note)**.** The companion note 'level-set-deficit-identity.md' (audit-trail markdown note, not a mathematical reference) wrote the second derivative as $I''(s) = -1/a(u^*(s))$ with $a(t) := \int_{\partial^* E_t} |\nabla u|$, i.e. with β in place of α. By Lemma 3.1 we have $\beta(t) = m(t)$, so this would give $I'' = -1/m$ — correct only when $|\nabla u|$ is constant on $\partial^* E_t$, i.e. on the ball, where Cauchy–Schwarz is sharp. The general identity is (8); the discrepancy between α and β is precisely the Cauchy–Schwarz defect D_H introduced below, and replacing α by β in I'' would silently delete D_H from the deficit identity. We use the α form throughout.

Cauchy–Schwarz. The two integrals $\alpha(t) = \int_{\partial^* E_t} |\nabla u|^{-1}\, d\mathcal{H}^{n-1}$ and $\beta(t) = \int_{\partial^* E_t} |\nabla u|\, d\mathcal{H}^{n-1}$ are linked by Cauchy–Schwarz on $\partial^* E_t$ applied to the pair $(|\nabla u|^{1/2}, |\nabla u|^{-1/2})$,

$$P(E_t)^2 = \Big(\int_{\partial^* E_t} 1\, d\mathcal{H}^{n-1}\Big)^2 \le \alpha(t)\beta(t), \tag{9}$$

with equality iff $|\nabla u|$ is $\mathcal{H}^{n-1}$-a.e. constant on $\partial^* E_t$.

5 Endpoint convexity gap and the deficit identity

Following the Nicola–Tilli convexity-gap idea adapted to torsion, introduce

$$G(s) := I(s) + \frac{s^{1+2/n}}{(4+2n)\,\omega_n^{2/n}}, \qquad s \in [0, |\Omega|]. \tag{10}$$

The second derivative of the polynomial corrector is computed directly: $\frac{d^2}{ds^2}\frac{s^{1+2/n}}{(4+2n)\omega_n^{2/n}} = \frac{1}{c_n s^{1-2/n}}$, with $c_n = n^2\omega_n^{2/n}$.

Lemma 5.1 (Pointwise convexity formula)**.** *For a.e. $s \in (0, |\Omega|)$, with $t = u^*(s)$,*

$$G''(s) = \frac{1}{c_n\, s^{1-2/n}} - \frac{1}{\alpha(t)}. \tag{11}$$

Proof. Lemmas 4.2, 4.3 give $G'' = I'' + \frac{1}{c_n s^{1-2/n}} = -1/\alpha(t) + \frac{1}{c_n s^{1-2/n}}$, which is (11). □

Definition 5.2 (Two pointwise defects)**.** For a.e. $t \in (0, \|u\|_\infty)$ set

$$D_H(t) := \alpha(t)\,\beta(t) - P(E_t)^2 = \Big(\int_{\partial^* E_t} \frac{1}{|\nabla u|}\Big)\Big(\int_{\partial^* E_t} |\nabla u|\Big) - P(E_t)^2, \tag{12}$$

$$D_I(t) := P(E_t)^2 - c_n\, m(t)^{2-2/n}. \tag{13}$$

By (9), $D_H(t) \ge 0$, with equality iff $|\nabla u|$ is constant on $\partial^* E_t$. By the De Giorgi isoperimetric inequality [5], $P(E_t) \ge n\omega_n^{1/n}|E_t|^{1-1/n}$ with equality iff E_t is (equivalent to) a ball, hence $D_I(t) \ge 0$ with equality iff E_t is a ball.

Lemma 5.3 (Pointwise-to-integrated convexity identity)**.** *With $L := |\Omega|$,*

$$\int_0^L s\, G''(s)\, ds = \int_0^{\|u\|_\infty} \frac{D_H(t) + D_I(t)}{c_n\, m(t)^{1-2/n}}\, dt. \tag{14}$$

Proof. From (11),

$$\int_0^L s\, G''(s)\, ds = \int_0^L s\Big(\frac{1}{c_n s^{1-2/n}} - \frac{1}{\alpha(u^*(s))}\Big) ds.$$

Change variable $s = m(t)$, $ds = m'(t)\, dt = -\alpha(t)\, dt$ on the strictly-decreasing branch of m (the at most countable plateau levels of u are an $\mathcal{L}^1$-null set in t, contributing zero):

$$\int_0^L s\, G''(s)\, ds = \int_0^{\|u\|_\infty} m(t)\Big(\frac{1}{c_n m(t)^{1-2/n}} - \frac{1}{\alpha(t)}\Big)\alpha(t)\, dt = \int_0^{\|u\|_\infty} \Big(\frac{m(t)\alpha(t)}{c_n m(t)^{1-2/n}} - m(t)\Big) dt.$$

Use $m(t) = \beta(t)$ (Lemma 3.1) and add and subtract $P(E_t)^2/(c_n m(t)^{1-2/n})$ in the integrand:

$$\frac{m(t)\alpha(t)}{c_n m(t)^{1-2/n}} - m(t) = \frac{\alpha(t)\beta(t) - c_n m(t)^{2-2/n}}{c_n m(t)^{1-2/n}} = \frac{D_H(t) + D_I(t)}{c_n m(t)^{1-2/n}}.$$

Substituting yields (14). □

Endpoint identification. With $L := |\Omega|$ define

$$\Gamma(\Omega) := G'(L) - \frac{G(L)}{L}. \tag{15}$$

Lemma 5.4 (Endpoint convexity-gap formula)**.** $\Gamma(\Omega) = \frac{1}{L}\int_0^L s\,G''(s)\,ds$.

Proof. $G \in W^{2,1}_{\mathrm{loc}}((0,L])$ and $G(0) = I(0) = 0$. Integration by parts on (ε, L):

$$\int_\varepsilon^L s\,G''(s)\,ds = \big[s\,G'(s)\big]_\varepsilon^L - \int_\varepsilon^L G'(s)\,ds = L\,G'(L) - \varepsilon\,G'(\varepsilon) - G(L) + G(\varepsilon).$$

By Lemma 4.2, $G'(s) = u^*(s) + \frac{s^{2/n}}{2n\omega_n^{2/n}}$ (using $\frac{1+2/n}{4+2n} = \frac{1}{2n}$), which is bounded; hence $\varepsilon G'(\varepsilon) \to 0$ and $G(\varepsilon) \to G(0) = 0$ as $\varepsilon \downarrow 0$. Dividing by L gives the claim. □

Lemma 5.5 (Endpoint value)**.**

$$\Gamma(\Omega) = \frac{2}{|\Omega|}\big(E(\Omega) - E(B)\big). \tag{16}$$

Proof. At $s = L$, $u^*(L) = 0$. By Lemma 4.1, $I(L) = \int_0^L u^* = \int_\Omega u\,dx$, so

$$\frac{G(L)}{L} = \frac{1}{L}\int_\Omega u\,dx + \frac{L^{2/n}}{(4+2n)\omega_n^{2/n}}.$$

By Lemma 4.2, $I'(L) = u^*(L) = 0$, so $G'(L) = \frac{L^{2/n}}{2n\omega_n^{2/n}}$. Hence

$$\Gamma(\Omega) = \frac{L^{2/n}}{2n\omega_n^{2/n}} - \frac{L^{2/n}}{(4+2n)\omega_n^{2/n}} - \frac{1}{L}\int_\Omega u\,dx = \frac{L^{2/n}}{2n(n+2)\omega_n^{2/n}} - \frac{1}{L}\int_\Omega u\,dx.$$

(Note $4+2n = 2(n+2)$, reconciling the constant $1/(4+2n)$ in the corrector (10) with $1/(2n(n+2))$ here.) For the ball $B = B_R$ with $|B_R| = L$, the torsion function is $u_B(x) = (R^2 - |x|^2)/(2n)$ (so $-\Delta u_B = 1$, $u_B = 0$ on ∂B_R). Take $|B| = \omega_n$ (so $R = 1$, $L = \omega_n$, $L^{2/n} = \omega_n^{2/n}$); direct integration in polar coordinates yields

$$\int_{B_1} u_B\,dx = \frac{1}{2n}\int_{B_1}(1 - |x|^2)\,dx = \frac{1}{2n}\cdot n\omega_n\int_0^1 (1 - r^2)r^{n-1}\,dr = \frac{\omega_n}{2n(n+2)}.$$

Substituting into Γ (still with $L = \omega_n$), $\frac{1}{L}\int_B u_B = \frac{1}{2n(n+2)}$ and $\frac{L^{2/n}}{2n(n+2)\omega_n^{2/n}} = \frac{1}{2n(n+2)}$, hence $\Gamma(B) = 0$. By scaling, the same conclusion holds for arbitrary $L > 0$, giving $\frac{L^{2/n}}{2n(n+2)\omega_n^{2/n}} = \frac{1}{L}\int_B u_B\,dx$. Subtracting,

$$\Gamma(\Omega) = \frac{1}{L}\Big(\int_B u_B\,dx - \int_\Omega u\,dx\Big) = \frac{2}{L}\big(E(\Omega) - E(B)\big),$$

using $E(\Omega) = -\frac{1}{2}\int_\Omega u$. The identification of $\Gamma(B)$ with the radial computation is the classical Pólya–Szegő content [12], [2, §II.5]. □

Theorem 5.6 (Exact deficit identity on reduced-boundary level sets)**.** *Let $\Omega \subset \mathbb{R}^n$ be open with $|\Omega| < \infty$ and $u \in H_0^1(\Omega)$ the variational torsion function. Then for a.e. $t \in (0, \|u\|_\infty)$, the set $E_t = \{u > t\}$ has finite perimeter, the identities* (3), (4), (8) *hold, and*

$$\boxed{\frac{2}{|\Omega|}\big(E(\Omega) - E(B)\big) = \frac{1}{|\Omega|}\int_0^{\|u\|_\infty} \frac{D_H(t) + D_I(t)}{n^2\omega_n^{2/n}\,m(t)^{1-2/n}}\,dt,} \tag{17}$$

where D_H, D_I are defined in (12)–(13)*. Both $D_H(t) \geq 0$ and $D_I(t) \geq 0$ a.e.*

Proof. Combine Lemmas 5.3, 5.4, 5.5. □

6 Cauchy–Schwarz / isoperimetric interpretation

The two defects D_H and D_I are exactly the two convexity gaps combined by the Pólya–Szegő symmetrization of the spectral profile.

- By Cauchy–Schwarz on $\partial^* E_t$ applied to $(|\nabla u|^{1/2}, |\nabla u|^{-1/2})$, $D_H(t) = \alpha(t)\beta(t) - P(E_t)^2 \geq 0$ with equality iff $|\nabla u|$ is $\mathcal{H}^{n-1}$-a.e. constant on $\partial^* E_t$. This is a level-set Serrin condition: it forces the gradient of the torsion function to be constant on each (a.e.) reduced-boundary level surface.
- By the De Giorgi isoperimetric inequality $P(E_t) \geq n\omega_n^{1/n}|E_t|^{1-1/n}$ [5, 11], squaring gives $D_I(t) = P(E_t)^2 - c_n m(t)^{2-2/n} \geq 0$, with equality iff E_t is (equivalent to) a ball.
- Their weighted sum, integrated against $dt/(c_n m(t)^{1-2/n})$, equals $\frac{2}{|\Omega|}(E(\Omega) - E(B))$. Rigidity in (17) therefore forces, simultaneously and for a.e. $t \in (0, \|u\|_\infty)$, both an overdetermined Serrin-type condition ($|\nabla u|$ level-constant) and sphericality of $\partial^* E_t$, i.e. the Saint–Venant rigidity of the ball.

Remark 6.1 (Weighted variance form of D_H)**.** Writing $f := |\nabla u|$ on $\partial^* E_t$ and $\bar{f} := m(t)/P(E_t)$,

$$\int_{\partial^* E_t} \frac{(f - \bar{f})^2}{f}\, d\mathcal{H}^{n-1} = \frac{m(t)}{P(E_t)^2}\, D_H(t),$$

so small $D_H(t)$ says that $|\nabla u|$ is almost constant on $\partial^* E_t$ in the natural weighted $L^2(1/f)$ sense [4]. This is the precise bridge to the overdetermined Serrin variance defect used downstream [16, 17].

7 Gradient regularity on level surfaces and the unit-weight variance bound

The downstream metric closure [17] requires a pointwise control of the *asymmetric-velocity defect* $\left(\int_{\partial^* E_t} |V_\rho - \bar{V}_\rho|\, d\mathcal{H}^{n-1}\right)^2$ by $D_H(t)$, where $V_\rho(x) := -t'(\rho)/|\nabla u(x)|$ is the normal velocity of $\partial^* E_\rho$ in ρ-parametrisation and $\bar{V}_\rho := P(B_\rho)/P(E_\rho)$ is the spatial mean of V_ρ. Unlike the weighted variance of Remark 6.1, which controls the $L^2(d\mathcal{H}/f)$-norm of $f - \bar{f}$, the asymmetric-velocity defect is the L^1-norm of $\phi - \bar{\phi}$ where $\phi := 1/f$. Converting between these requires a uniform two-sided bound on $f = |\nabla u|$ on the level surface.

7.1 The standing gradient hypothesis (Hyp-G)

Hypothesis 7.1 (Two-sided gradient bound on level surfaces)**.** There exist constants $0 < m_0 \leq M_0 < \infty$ depending only on n, R, ρ_* such that, for $\mathcal{L}^1$-a.e. $\rho \in [\rho_*, \rho_\delta]$,

$$m_0 \;\leq\; |\nabla u(x)| \;\leq\; M_0 \qquad \text{for } \mathcal{H}^{n-1}\text{-a.e. } x \in \partial^* E_\rho.$$

Remark 7.2 (Validity of Hyp-G in the bounded reduction class)**.** Hypothesis 7.1 is a working assumption of the metric closure of [17], valid in the bounded reduction class $\Omega \subset B_{R_*(n)}$ produced by [19, Lemma 5.3]. The justification has two parts.

Upper bound $M_0 = M_0(n, R)$ — unconditional in the bounded reduction class. The variational torsion function $u \in W_0^{1,2}(\Omega)$ satisfies $-\Delta u = 1$ in Ω with $\Omega \subset B_R$ open of finite volume. For the bounded reduction class [19, Lemma 5.3], which produces Ω with a uniform boundary regularity (in particular Ω John-domain or Lipschitz-equivalent property inherited from the BDV construction), the global gradient bound

$$\|\nabla u\|_{L^\infty(\Omega)} \;\leq\; K_n\, R,$$

holds with K_n depending only on n. Two routes give this: (i) Talenti's rearrangement [13] produces the pointwise profile bound $|\nabla u^*|(s) \le \rho/n$ on the symmetrised problem, which transfers to $|\nabla u| \le K_n R$ pointwise in Ω via the boundary regularity of the reduction class (e.g. via barrier arguments at $\partial\Omega$); (ii) the global gradient L^∞ regularity theory of Cianchi–Maz'ya [9, Thm. 1.1, applied to the linear case $p = 2$] covers the same statement directly, with Ω of finite Lebesgue measure and L^q source for $q > n$ (both holding for $f \equiv 1$ in $\Omega \subset B_R$). Thus the upper bound $M_0 := K_n R$ is unconditional in the bounded reduction class.

Lower bound $m_0 = m_0(n, \rho_*)$ — the residual conditionality. Brothers–Ziemer non-degeneracy [4, Thm. 1] establishes $|\nabla u| > 0$ $\mathcal{H}^{n-1}$-a.e. on $\partial^* E_\rho$ for a.e. ρ, ruling out plateau levels. Talenti's gradient comparison [13] provides the *rearranged* bound $|\nabla u^*|(s) \le |\nabla u_B^*|(s)$ where u_B is the ball torsion with $|B| = |\Omega|$, and the ball profile $|\nabla u_B^*|(\omega_n \rho^n) = \rho/n$ gives a quantitative *upper* bound on $|\nabla u^*|$. The corresponding quantitative *lower* bound $|\nabla u(x)| \ge m_0(n, \rho_*)$ $\mathcal{H}^{n-1}$-a.e. on $\partial^* E_\rho$, uniform in $\rho \in [\rho_*, \rho_\delta]$, does *not* follow from these classical inputs alone. In the small-deficit regime $\delta_T \le \delta_0$ it is expected from stability considerations (the level set $\partial^* E_\rho$ is close to ∂B_ρ where $|\nabla u_B| = \rho/n \ge \rho_*/n > 0$, and a perturbation argument should transfer this); a proof would go through a Caffarelli-type non-degeneracy estimate on annular shells inside $E_\rho \setminus E_{\rho+\varepsilon}$ [10, §7], exploiting the explicit non-zero driving term $-\Delta u = 1$.

We adopt the lower bound part of Hyp-G as a *standing hypothesis* throughout the metric closure. Verifying this hypothesis is the residual conditionality of the unconditional sharp Faber–Krahn inequality; the rest of the chain is rigorous given the hypothesis. The constants m_0, M_0 enter the downstream chain only as $(M_0/m_0)^2$ in the constant C_{vel} of Proposition 7.3.

7.2 The unit-weight variance bound

Proposition 7.3 (Velocity defect from D_H under Hyp-G)**.** *Under Hypothesis 7.1, for $\mathcal{L}^1$-a.e. $\rho \in [\rho_*, \rho_\delta]$,*

$$\left(\int_{\partial^* E_\rho} |V_\rho(x) - \bar{V}_\rho| \, d\mathcal{H}^{n-1}(x) \right)^2 \le \frac{(M_0/m_0)^2}{n^2} \, D_H(t(\rho)). \tag{18}$$

Equivalently, $\left(\int |V_\rho - \bar{V}_\rho|\right)^2 \le C_{\mathrm{G}}(n, R, \rho_*) \, D_H(t(\rho))$ with $C_{\mathrm{G}} := (M_0/m_0)^2/n^2$.

Proof. Set $\phi(x) := 1/f(x)$ with $f := |\nabla u|$, and $\bar{\phi} := \alpha(t)/P(E_t)$ where $\alpha(t) := \int_{\partial^* E_t} 1/f \, d\mathcal{H}^{n-1}$. Then $\int (\phi - \bar{\phi}) \, d\mathcal{H}^{n-1} = 0$, so $\phi - \bar{\phi}$ has $\mathcal{H}^{n-1}$-mean zero on $\partial^* E_t$.

Step 1: Cauchy–Schwarz. By $\int |\phi - \bar{\phi}| \cdot 1 \, d\mathcal{H}^{n-1} \le \sqrt{P(E_t)} \sqrt{\int (\phi - \bar{\phi})^2 \, d\mathcal{H}^{n-1}}$,

$$\left(\int |\phi - \bar{\phi}| d\mathcal{H}^{n-1} \right)^2 \le P(E_t) \int (\phi - \bar{\phi})^2 d\mathcal{H}^{n-1}. \tag{19}$$

Step 2: Variance expansion. $\int (\phi - \bar{\phi})^2 = \int \phi^2 - 2\bar{\phi} \int \phi + \bar{\phi}^2 P = \int 1/f^2 - \alpha^2/P$, so

$$P(E_t) \int (\phi - \bar{\phi})^2 \, d\mathcal{H}^{n-1} = P(E_t) \int 1/f^2 \, d\mathcal{H}^{n-1} - \alpha(t)^2. \tag{20}$$

Step 3: Symmetrised double-integral forms. By the elementary symmetrising identity on $\partial^* E_t \times \partial^* E_t$,

$$P(E_t) \int 1/f^2 d\mathcal{H}^{n-1} - \alpha(t)^2 = \tfrac{1}{2} \iint \frac{(f(x) - f(y))^2}{f(x)^2 f(y)^2} d\mathcal{H}_x^{n-1} d\mathcal{H}_y^{n-1}. \tag{21}$$

Indeed, expanding the right-hand side and exchanging the labels x, y in each cross-term yields $\iint 1/f(x)^2 - \iint 1/(f(x)f(y)) = P \int 1/f^2 - \alpha^2$, and the symmetrisation halves the result. An identical computation applied to $D_H(t) = \alpha(t)\beta(t) - P(E_t)^2 = \int 1/f \int f - (\int 1)^2$ gives

$$D_H(t) = \tfrac{1}{2} \iint \frac{(f(x) - f(y))^2}{f(x) f(y)} d\mathcal{H}_x^{n-1} d\mathcal{H}_y^{n-1}. \tag{22}$$

(Equation (22) is the standard Cauchy–Schwarz defect formula for the pair $(\sqrt{f}, 1/\sqrt{f})$; cf. [4, §1].)

Step 4: Pointwise comparison via Hyp-G. The integrand of (21) equals $1/(f(x)f(y))$ times the integrand of (22). By Hypothesis 7.1, $f(x), f(y) \geq m_0$, so $1/(f(x)f(y)) \leq 1/m_0^2$, and

$$P(E_t)\int 1/f^2 d\mathcal{H}^{n-1} - \alpha(t)^2 \;\leq\; \frac{1}{m_0^2} D_H(t). \tag{23}$$

Step 5: Talenti $|t'| \leq 1/n$. The volume-radius parametrisation $|E_\rho| = \omega_n \rho^n =: s$ gives $s = \omega_n\rho^n$ and $t(\rho) = u^*(s)$ with u^* the Schwarz decreasing rearrangement of u. Chain rule: $t'(\rho) = (u^*)'(s){\cdot}n\omega_n\rho^{n-1}$. For the ball torsion $u_B = (1-|x|^2)/(2n)$ with $|B| = \omega_n$, the rearranged profile is $u_B^*(s) = (1-(s/\omega_n)^{2/n})/(2n)$, and $(u_B^*)'(s) = -s^{2/n-1}/(n^2\omega_n^{2/n})$, so $t_B'(\rho) = -\rho/n$. Talenti's pointwise gradient comparison $|(u^*)'(s)| \leq |(u_B^*)'(s)|$ [13, Thm. 1] then gives

$$|t'(\rho)| \;\leq\; |t_B'(\rho)| \;=\; \rho/n \;\leq\; 1/n \qquad (\rho \in (0,1]).$$

Step 6: Conclusion. Combining (19), (20), (23) and Step 5,

$$\Big(\int |V_\rho - \bar{V}_\rho|\Big)^2 = t'(\rho)^2\Big(\int |\phi - \bar{\phi}|\Big)^2 \leq \frac{1}{n^2}\cdot\frac{D_H(t(\rho))}{m_0^2}.$$

This is the bare quantitative content of the proof. For notational uniformity with downstream blocks that also invoke the upper bound M_0 (e.g. the perimeter/L^∞-gradient estimates of [17]) we record the bound in the form

$$\Big(\int |V_\rho - \bar{V}_\rho|\Big)^2 \leq \frac{C_{\mathrm{vel}}}{n^2} D_H(t(\rho)), \qquad C_{\mathrm{vel}} := \max(1, M_0^2)/m_0^2,$$

since $\max(1, M_0^2) \geq 1$ ensures $C_{\mathrm{vel}} \geq 1/m_0^2$; in the bounded reduction class $M_0 = M_0(n,R) \geq K_nR \geq K_n{\cdot}2 \geq 1$ (with $R \geq 2$), so $C_{\mathrm{vel}} = (M_0/m_0)^2$ in practice and the displayed inequality (18) holds with this constant. □

Corollary 7.4 (Integrated form)**.** *Under Hypothesis 7.1, with $d\mu(\rho) = -t'(\rho)d\rho$,*

$$\int_{\rho_*}^{\rho_\delta}\Big(\int_{\partial^* E_\rho}|V_\rho - \bar{V}_\rho| d\mathcal{H}^{n-1}\Big)^2 d\mu(\rho) \;\leq\; C_{\mathrm{vel}}(n, R, \rho_*)\,\delta_T(\Omega),$$

where $C_{\mathrm{vel}} := c_{n,\rho_*}(M_0/m_0)^2$ with the constant c_{n,ρ_*} depending only on n and ρ_*.

Proof. Apply Proposition 7.3 and integrate against $d\mu$: $\int_{\rho_*}^{\rho_\delta}\big(\int |V - \bar{V}|\big)^2 d\mu \leq (M_0/m_0)^2 n^{-2}\int_{\rho_*}^{\rho_\delta} D_H(t(\rho))\,d\mu(\rho)$. By the deficit identity (17) of Theorem 5.6, with the weight $1/(c_n m(t)^{1-2/n}) \geq c'_{n,\rho_*} > 0$ on $[\rho_*, \rho_\delta]$,

$$\int_{\rho_*}^{\rho_\delta} D_H(t(\rho))\,d\mu(\rho) = \int_{t(\rho_\delta)}^{t(\rho_*)} D_H(t)\,dt \;\leq\; c_{n,\rho_*}\,|\Omega|\,(E(\Omega) - E(B)) = c'_{n,\rho_*}\,\delta_T(\Omega),$$

where in the last step we used $|\Omega| = \omega_n$ and the normalisation $\delta_T(\Omega) := E(\Omega) - E(B^*) \geq 0$ (the proportionality constant absorbs the factor 2 in (17)). □

8 Discussion and failure modes

We collect the regularity assumptions that are *not* used and the ones that are, since the latter is the precise input the downstream blocks rely on.

1. Ω *need not be bounded.* The only consequence of finite measure is the Talenti L^∞ bound (1), which is enough to make u^* bounded and to make $\beta \in L^1$ on $(0, \|u\|_\infty)$ via the Dirichlet identity. Boundedness $\Omega \subset B_R$ enters only later in the metric closure [14, 17], where strong isoperimetry with radial trace control is used.

2. Ω *need not be smooth.* Lemmas 2.1 and 3.1 use BV coarea and Lipschitz truncation; neither requires $\partial\Omega$ to be a manifold. There is no need for a one-sided trace of ∇u on $\partial^* E_t$ (cf. Remark 3.2).

3. Ω *need not be connected.* The torsion function decouples across connected components and the identity is additive in the obvious sense (the dt integral pools all components).

4. *The identity is a.e., not pointwise.* Lemmas 2.1, 3.1, 4.3 hold for t outside an $\mathcal{L}^1$-null set of bad levels: at most countably many plateau levels of u (where $|\{u = t\}| > 0$), the Sobolev–Sard exceptional set, and BV-coarea exceptional levels. Levels in this null set are absorbed by the dt integral.

5. *All integrals are on reduced boundaries.* The cusps, self-tangencies, and the unreduced part of ∂E_t are $\mathcal{H}^{n-1}$-null on the complement of $\partial^* E_t$ at a.e. level [5, 11], hence contribute nothing. No graph parametrization of $\partial^* E_t$ is used or available in general.

6. *The corrected I''.* The use of α rather than β in Lemma 4.3 is essential. The naive guess $I'' = -1/\beta = -1/m$ holds only on the ball, where $|\nabla u|$ is level-constant and Cauchy–Schwarz is sharp; in general it would erase D_H from the identity (17). See Remark 4.4.

Forward references. Theorem 5.6 is the input identity for the rest of the Plan 2 chain. The metric framework [14] extracts a first-variation/velocity defect from the integrated $D_H + D_I$. The weighted metric trace [17] uses the levelwise Fusco–Julin centre and a good/bad isoperimetric split to promote integrated action to a pointwise endpoint at $\widehat{\rho}$. The boundary-layer transfer [18] carries control on $E_{\widehat{\rho}}$ back to Ω via Lemma 8.3 of the source note, and the global assembly [19] closes the chain with bounded reduction and the Kohler–Jobin transfer. The centroid and space–time blocks [15, 16] remain auxiliary diagnostics for the alternate centroid route.

What is genuinely new vs. cited. The BV coarea, the divergence theorem on sets of finite perimeter, the rearrangement framework, and the Pólya–Szegő identification of $\Gamma(B) = 0$ are all standard [11, 1, 13, 7, 12, 2]. The novel contributions are: (a) the combined finite-perimeter statement of the deficit identity on reduced boundaries for a.e. level, written without any selection or graph-entry assumption; (b) the Lipschitz-truncation proof of Lemma 3.1, replacing the formal trace $\nabla u \cdot \nu_{E_t} = -|\nabla u|$; and (c) the corrected $I'' = -1/\alpha$ in Lemma 4.3, required for the deficit identity to hold off the ball.

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Companion blocks (Plan 2 internal cross-references). The following are internal cross-references to sibling Plan 2 building blocks, not external bibliography.

[14] *Metric Framework: $\mathcal{X}$, first variation, per-ρ velocity defect*, Plan 2 building block MetricFramework (Plan 2 internal).

[15] *H^1 centroid bound*, Plan 2 building block CentroidBound (Plan 2 internal).

[16] *Space–time Π identity and (B^2-int) via slicing-then-squaring*, Plan 2 building block SpaceTimeIdentity (Plan 2 internal).

[17] *Integrated action to pointwise endpoint at $\widehat{\rho}$*, Plan 2 building block WeightedMetricTrace (Plan 2 internal).

[18] *Boundary-layer transfer: $E_{\widehat{\rho}} \to \Omega$, bounded sharp Saint–Venant*, Plan 2 building block BoundaryLayerTransfer (Plan 2 internal).

[19] *Bounded reduction and Kohler–Jobin transfer to sharp Faber–Krahn*, Plan 2 building block GlobalAssembly (Plan 2 internal).